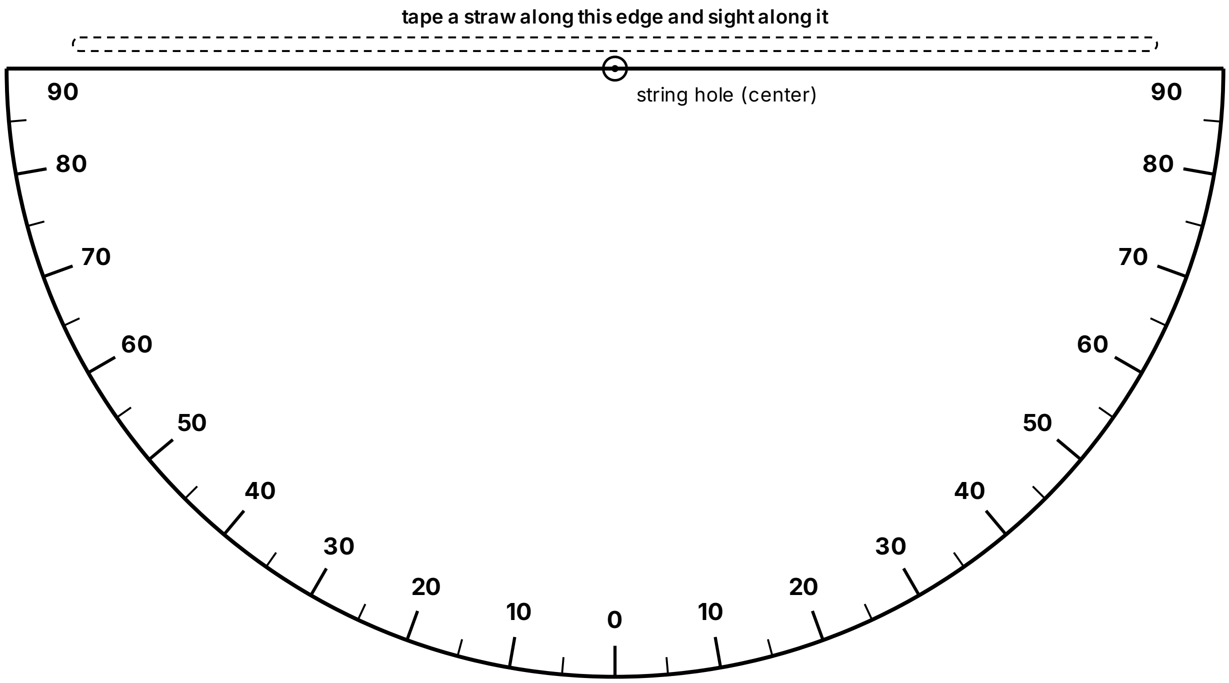
Paper clinometer and tan table

Read the angle to the treetop, then use the tan table (no calculator needed) to find the height.



Cut along the flat top edge and the curved outline.

ASSEMBLE

01 Cut out the half circle along the curved outline.

02 Tape a straight straw along the top straight edge.

03 Punch the center dot, thread a string through it, and tie a washer on the end as a weight.

04 Sight the treetop through the straw, let the string hang, then pinch it against the scale and read the angle.

05 Self-check: sight something level (reads 0) and straight up (reads 90).

FIND THE TREE HEIGHT

Tree height = eye height + (distance to the tree × tan of the angle). Use the standoff distance the instructor pre-marked with the long tape.

ANGLE	TAN
15°	0.27
20°	0.36
25°	0.47
30°	0.58
35°	0.70
40°	0.84
45°	1.00

ANGLE	TAN
50°	1.19
55°	1.43
60°	1.73
65°	2.14
70°	2.75
75°	3.73

Example: eye height 1.5 m, distance 15 m, angle 40° gives 1.5 + 15 × 0.84 = about 14.1 m.